

STA304XS_Assignment_1

Saurav Sathnarayan

2025-08-21

Question 1

We aim to find the posterior distribution using the following formula:

$$\pi(\theta|V) \propto L(\theta, V) \pi(\theta)$$

We know that $\pi(\theta) \propto 1$, thus our posterior distribution is just

$$\pi(\theta|V) \propto L(\theta, V)$$

From our model equation V is multi-variate normally distributed, $V \sim \mathcal{N}_2(aMv_{t-1}, I_2)$

Our pdf is:

$$f(V) = \frac{1}{\sqrt{2\pi}} e^{(-\frac{1}{2}(v_t - aMv_{t-1})^T(v_t - aMv_{t-1}))}$$

We can simplify to make it easier for further calculations

$$f(V) \propto e^{(-\frac{1}{2}(v_t - aMv_{t-1})^T(v_t - aMv_{t-1}))}$$

Focusing on the exponent:

$$(v_t - aMv_{t-1})^T(v_t - aMv_{t-1}) = v_t^T v_t - v_t^T 2aMv_{t-1} + a^2 v_{t-1}^T M^T M v_{t-1} = v_t^T v_t - v_t^T 2aMv_{t-1} + a^2 v_{t-1}^T v_{t-1}$$

The terms $v_t^T v_t$ and $a^2 v_{t-1}^T v_{t-1}$ are independent of θ thus we can factor it out of the pdf to get:

$$f(V) \propto e^{(av_t^T M v_{t-1})}$$

We can further simplify our exponent:

$$v_t^T M v_{t-1} = \begin{pmatrix} v_{t_1} & v_{t_2} \end{pmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{pmatrix} v_{t-1_1} \\ v_{t-1_2} \end{pmatrix} = \cos \theta (v_{t_1} v_{t-1_1} + v_{t_2} v_{t-1_2}) + \sin \theta (v_{t_1} v_{t-1_2} - v_{t_2} v_{t-1_1})$$

So we end up with:

$$f(V) \propto \exp \frac{\cos \theta (v_{t_1} v_{t-1_1} + v_{t_2} v_{t-1_2}) + \sin \theta (v_{t_1} v_{t-1_2} - v_{t_2} v_{t-1_1})}{2}$$

Our likelihood is simply:

$$L(\theta, V) = \prod_{t=1}^n f(V, \theta)$$

$$L(\theta, V) \propto \exp \frac{\sum_{t=1}^n (\cos \theta (v_{t1} v_{t1,1} + v_{t2} v_{t-1,2}) + \sin \theta (v_{t1} v_{t-1,2} - v_{t2} v_{t-1,1}))}{2}$$

Distributing our summation we get $s_1 = \sum (v_{t1} v_{t-1,1} + v_{t2} v_{t-1,2})$ and $s_2 = \sum (v_{t1} v_{t-1,2} - v_{t2} v_{t-1,1})$

Therefore:

$$\pi(\theta | V) \propto \exp\left(\frac{s_1 \cos \theta + s_2 \sin \theta}{2}\right)$$

Question 2

- (a) We aim to sample from the posterior distribution of the turning angle, $\pi(\theta | V)$, which is proportional to $\exp((s_1 \cos \theta + s_2 \sin \theta)/2)$ for $-\pi < \theta < \pi$. Since this distribution is not standard, we use the **Acceptance-Rejection algorithm**. A candidate θ^* is drawn from a proposal distribution $h(\theta)$, and it is accepted with probability $\gamma(\theta^*)$, which is the ratio of the target density to a scaled proposal density. Repeating this procedure produces samples approximately from $\pi(\theta | V)$. Our formula for $\gamma(\theta)$ is:

$$\gamma(\theta) = \frac{\pi(\theta)g(t, \theta)}{Ch(\theta)}$$

Where $h(\theta) \propto 3 \cos(\theta)$

$$C = \max_{\theta} \frac{\pi(\theta)g(t, \theta)}{h(\theta)}$$

We can find our value for C using R's optimise function:

```
#posterior, same as likelihood, since prior is constant.
posterior <- function(theta) {
  exp((s1 * cos(theta) + s2 * sin(theta))/2)
}

#candidate
h <- function(theta) {
  exp(3 * cos(theta))
}

#gamma function
gamma <- function(theta, C) {
  (posterior(theta)/(h(theta)*C))
}

opt <- optimize(function(theta) posterior(theta)/h(theta),
  interval = c(-pi, pi), maximum = TRUE)
```

```
C <- opt$objective
```

```
C
```

```
## [1] 3.588122e+87
```

We can then use the acceptance- Rejection algorithm to create sample draws from our posterior as below:

```
# Acceptance-Rejection function
ar <- function(n= 5000) {
  samples <- numeric(0)

  while(length(samples) < n) {
    # Propose from candidate h(theta) ~ exp(3 cos(theta))
    # Simplest: uniform [-pi, pi] and weight via gamma
    theta_star <- runif(1, -pi, pi)
    u <- runif(1)
    g <- gamma(theta_star,C)
    if(u < g) {
      samples <- c(samples, theta_star)
    }
  }

  return(samples)
}

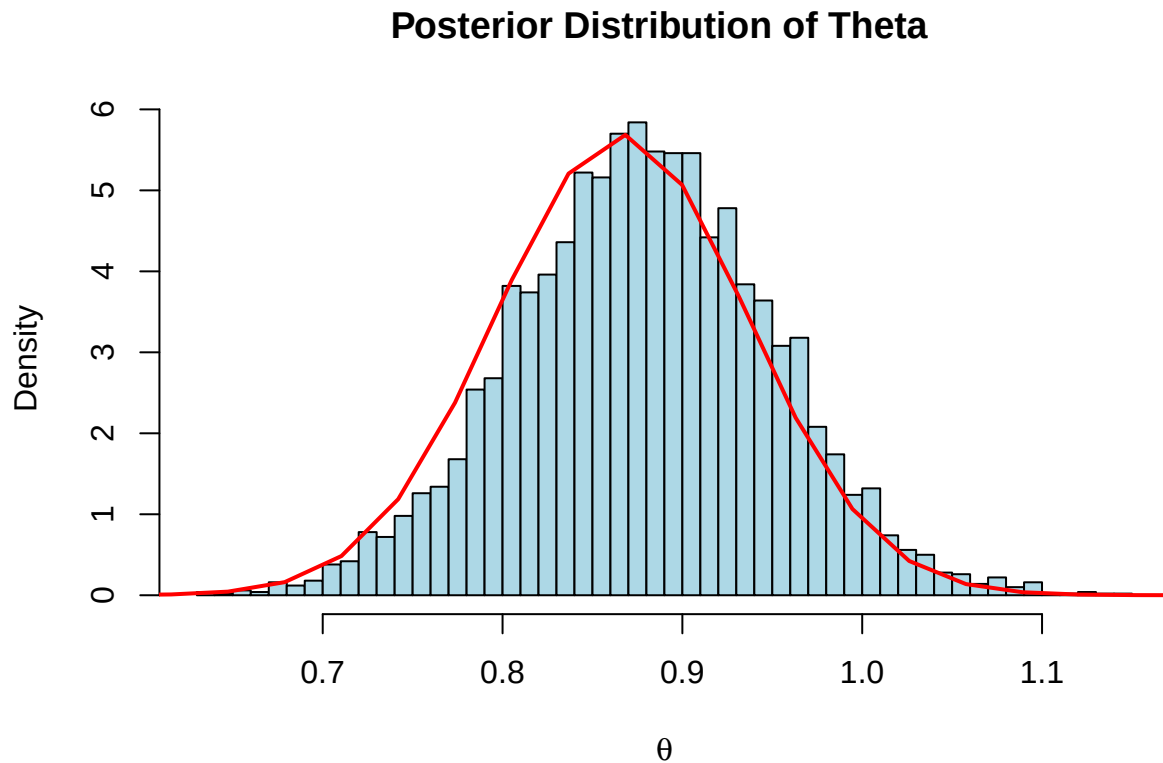
# Generate posterior draws
set.seed(123)
samples <- ar(n =5000)
```

(b)

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.6319 0.8302 0.8778 0.8781 0.9262 1.1404
```

```
## Posterior mean: 0.8781108
```

```
## Posterior sd: 0.07169482
```



When plotting posterior samples using `hist(..., probability = TRUE)` in R, the histogram is scaled so that the area under the curve sums to 1, approximating a probability density function. The height of the bars represents the density, not the probability of individual values. Therefore, if the data are concentrated in a narrow range or if the bin width is small, the density can exceed 1. This is mathematically valid because the integral (or sum of bar areas) still equals 1, and the density value itself is not restricted to be less than or equal to 1. It simply reflects how concentrated the distribution is over a small interval.

Question 3

Appendix

```
knitr::opts_chunk$set(echo = TRUE)
load("/Users/sauravsathnarayan/Desktop/University/STA304XS/STA304XS assignment 1/STA3043_Assignment1_2020.Rsave")
data <- as.data.frame(Class.List[["STHSAU001"]])
mu <- cbind(data$x,data$y)
vt <- rbind(c(0,0), diff(mu))
library(circular)

n <- nrow(vt)

v_t1 <- vt[2:n, 1]
v_t2 <- vt[2:n, 2]
v_tminus1_1 <- vt[1:(n-1), 1]
v_tminus1_2 <- vt[1:(n-1), 2]
```

```

s1 <- sum(v_t1 * v_tminus1_1 + v_t2 * v_tminus1_2)
s2 <- sum(v_t1 * v_tminus1_2 - v_t2 * v_tminus1_1)

#posterior, same as likelihood, since prior is constant.
posterior <- function(theta) {
  exp((s1 * cos(theta) + s2 * sin(theta))/2)
}

#candidate
h <- function(theta) {
  exp(3 * cos(theta))
}

#gamma function
gamma <- function(theta, C) {
  (posterior(theta)/(h(theta)*C))
}
opt <- optimize(function(theta) posterior(theta)/h(theta),
  interval = c(-pi, pi), maximum = TRUE)
C <- opt$objective

C
# Acceptance-Rejection function
ar <- function(n= 5000) {
  samples <- numeric(0)

  while(length(samples) < n) {
    # Propose from candidate  $h(\theta) \sim \exp(3 \cos(\theta))$ 
    # Simplest: uniform  $[-\pi, \pi]$  and weight via gamma
    theta_star <- runif(1, -pi, pi)
    u <- runif(1)
    g <- gamma(theta_star, C)
    if(u < g) {
      samples <- c(samples, theta_star)
    }
  }

  return(samples)
}

# Generate posterior draws
set.seed(123)
samples <- ar(n =5000)

# Posterior summaries
summary(samples)
cat("Posterior mean:", mean(samples), "\n")
cat("Posterior sd:", sd(samples), "\n")

# Plot posterior
hist(samples, breaks= 50, freq = FALSE,
  col = "lightblue", main = "Posterior Distribution of Theta ",
  xlab = expression(theta))

```

```
# Overlay posterior function (rescaled)
theta_grid <- seq(-pi, pi, length.out = 200)
posterior_vals <- sapply(theta_grid,posterior)
posterior_vals <- posterior_vals / sum(posterior_vals * diff(theta_grid[1:2])) # rescale
lines(theta_grid, posterior_vals, col = "red", lwd = 2)
```